Remote-Sensing Scene Classification via Multistage Self-Guided Separation Network

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***Abstract*— In recent years, remote-sensing scene classification is one of the research hotspots and has played an important role in the field of intelligent interpretation of remote-sensing data. However, various complex objects and backgrounds form a variety of remote-sensing scenes through spatial combination and correlation, which brings great challenges to accurately classifying different scenes. Among them, the insufficient feature difference brought about the unbalanced change of background and target between interclass samples and the feature representation inconsistency caused by the difference of representation among the intraclass samples have become obstacles to effectively distinguishing different scene images. To address these issues, a multistage self-guided separation network (MGSNet) is proposed for remote-sensing scene classification. First of all, different from the previous work, it attempts to utilize the background information outside the effective target in the image as a decision aid through a target–background separation strategy to improve the distinguishability between target similarity-background difference samples. In addition, the diversity of feature concerns among different network branches is expanded through contrastive regularization (CR) to improve the separation of target–background information. Additionally, a self-guided network is proposed to find common features between intraclass samples and improve the consistency of feature representation. It combines the texture and morphological features of images to guide feature learning, effectively reducing the impact of intraclass differences. Extensive experimental results on three benchmarks demonstrate that MGSNet can achieve better classification performance compared to the state-of-the-art approaches.**

***Index Terms*— Remote sensing, scene classification, self-guided network, target–background separation strategy.**

I. Introduction

REMOTE-sensing scene classification is a research hotspot in the intelligent interpretation task of remote-sensing data, which aims to focus on high-level semantic information in scene images and classify them into corresponding scene categories, while providing scene-level data understanding and decision aids for many practical applications, such as land use and cover monitoring, urban development and planning, and natural disaster response [1], [2], [3], [4], [5]. In recent years, with the continuous progress of remote-sensing observation technology, the application scope of remote-sensing images has become more and more extensive, and how to make full use of the increasing number of remote-sensing images for intelligent earth observation becomes extra important [6], [7], [8], [9]. Therefore, a great deal of research has been performed in the last decades to achieve scene classification [10], [11]. Feature extraction plays an important role in the scene classification task, and the expressiveness of the extracted features also determines the performance of the classification. According to the types of extracted features, existing remote-sensing scene classification work can be roughly divided into two directions: handcrafted feature-based methods and feature learning-based methods [12], [13].

Early methods for scene classification are mainly based on handcrafted feature extraction. These methods take the professional knowledge of practitioners as a priori to extract features such as color, texture, shape, and spatial and spectral information from scene images for decision support of scene classification. Li et al. [14] proposed an object-oriented classification method, which combined improved color structure code (CSC) to classify high-resolution data. In [15], a global morphological texture descriptor was designed to explore the potential of multiscale texture descriptors in scene classification tasks. In [16], a unique invariant feature extraction method was proposed, these features exhibited invariance to image scale and rotation, and can robustly identify objects in clutter and occlusion. Dalal and Triggs [17] studied the importance of directional gradient histograms in effective target detection. However, it is worth noting that methods relying on handcrafted-feature have good results on images with the same texture structure or spatial distribution, but they are still limited when faced with images with complex scenes because artificially designed features can significantly affect the representational ability of image features. To make up for the limitations of handcrafted feature-based methods, automatic feature extraction from images has gradually become a new feature extraction method.

This has been followed by the rise of feature learning-based methods. Feature learning-based methods [18], [19] can learn the corresponding adaptive functions from the original

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pixel values or handcrafted features through sparse coding, autoencoder, and other means, to obtain a more appropriate image representation for scene classification [12]. Compared to methods based on handcrafted features, the difference lies in the automatic learning of relevant features, rather than relying on manually designed features for discrimination. However, since the above-mentioned methods do not utilize the corresponding label information, their feature extraction and learning capability are limited, and it is not conducive to further improvement of scene classification performance.

In recent years, with the development of artificial intelligence, deep-learning methods provide excellent performance and have made important progress in image classification, object recognition, and semantic segmentation [20], [21], [22], [23]. The method based on deep feature learning enables the model to self-learning more powerful, abstract, and meaningful features through the deep network architecture, avoiding the defects of manually designed features. At the same time, the introduction of label information enables the model to learn more accurate distinctions between classes and improve classification performance [24], [25], [26]. In [27], a multiinstance densely connected ConvNet was proposed, which treated scene classification as a multiinstance learning problem to further investigate local semantics. In [28], a new discriminant function was introduced to improve the training effect of the model. For this reason, in addition to the common classification loss, the metric learning regularization term was also introduced and applied to CNN features, making the model more distinctive. Liu et al. [29] constructed a two-branch multiscale convolutional neural network using a fixed-scale network and a variable-scale network to cope with the scale variation of the target in the image, in addition to adding a similarity measurement layer to ensure the similarity between the original image and the scaled image features. In [30], an enhanced attention module was designed to enhance the feature extraction and generalization capabilities of the deep neural networks to better recognize small objects in scenes with complex backgrounds. Li et al. [31] attempted to use a pretrained convolutional neural network model to reduce training time and proposed a fusion strategy to integrate the multilayer features of the model. In [32], the convolutional local attention module was embedded in all downsampling and residual blocks of the ResNet backbone to construct a multibranch local attention network, which placed the convolutional channel attention module and the local spatial attention module in parallel to obtained channel and spatial attention, respectively, which helped to emphasize the main targets in complex backgrounds and mentions the representational power of features.

Although the above works have tried to solve the problems in scene classification tasks from various perspectives, there are still some problems when faced with classification problems in actual complex scenes.

1) Feature confusion due to the difference of target–background imbalance changes. The similarity between backgrounds and targets in interclass images leads to easily confused and poorly discriminative extracted features, where the well-known issue of target similarity between interclass images hinders the effective discrimination of different scenes. For example, the similarity of the targets of the samples shown in Fig. 1(a) leads to a high degree of confusion between different scenes. Furthermore, scene labels are usually determined by internal key targets in previous methods, ignoring the decision supplement that background information can provide. For example, the difference in background information in Fig. 1(a) can provide more information supplements for distinguishing classes with similar targets.
2) Inconsistent feature representation due to the diversity of the intraclass sample. As shown in Fig. 1(c), in the case of nonuniform sample collection and a large number of disturbances, such as seasonal changes, lighting intensity, and regional variations, the representation of intraclass samples show diversity, which poses challenges to the robustness of the model.

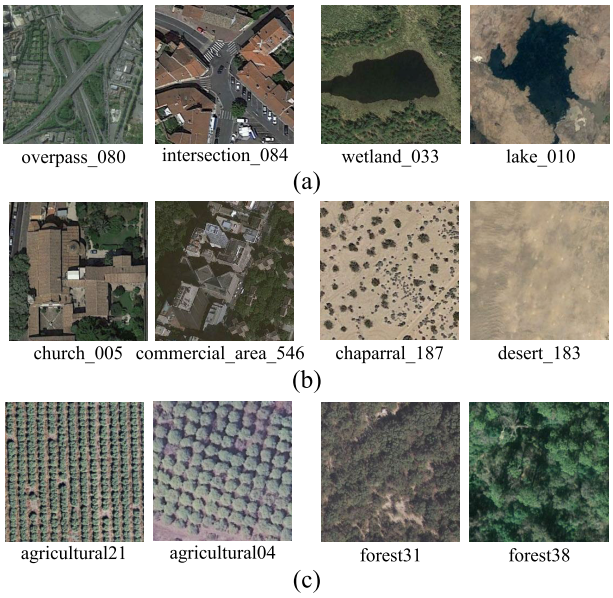


Fig. 1. Similarity of interclass images and the diversity of intraclass images. (a) Target similarity for interclass images. (b) Background similarity for interclass images. (c) Diversity for intraclass images.

To solve the above problems, a novel multistage self-guided separation network (MGSNet) is proposed for remote-sensing scene classification. First, to solve the problem of insufficient feature representation caused by the similarity due to the unbalanced change of target–background, attempts have been done to separate the background and target to extract corresponding features for the first time in the field of remote-sensing scene classification. Specifically, the target and background are distinguished by binary segmentation, followed by feature extraction of the target and background separately using a two-branch network. In addition, the differences between target features and background features are enlarged by introducing contrastive regularization (CR) [33], so that the key target features are fully mined while retaining the background information, providing assistance for better distinguishing scenes with similar targets. Second, to improve the feature consistency among intraclass samples with different representations, a self-guided network is developed, which

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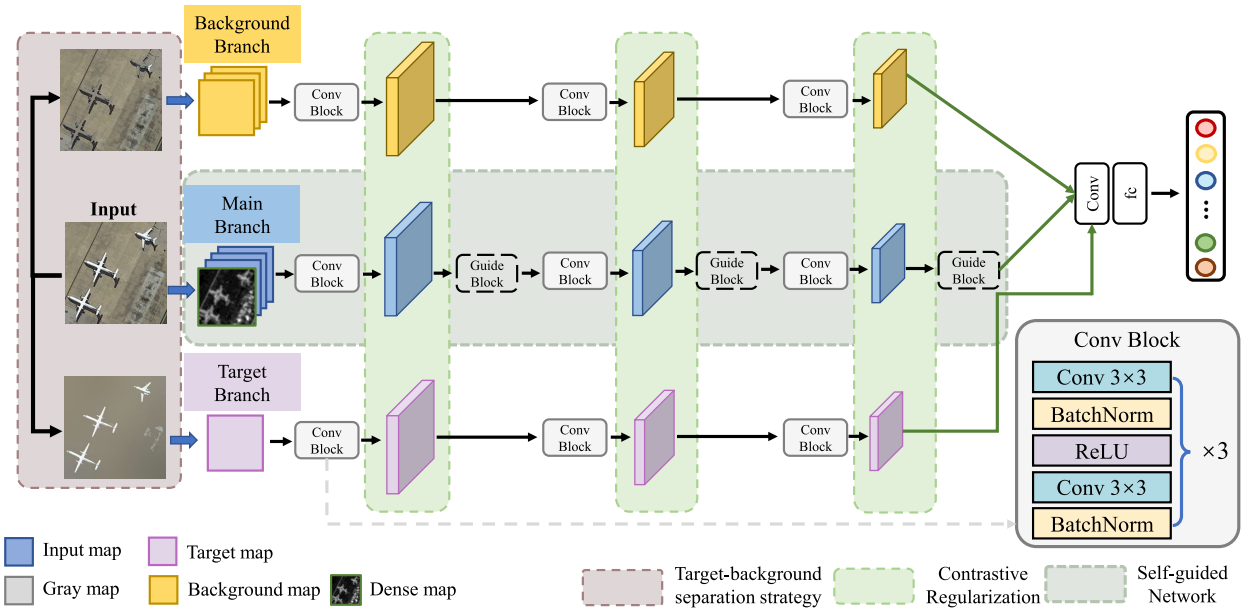


Fig. 2. Schematic illustration of the proposed MGSNet. First, for input images, a target–background separation strategy is applied to extract the target and background information in the scene and send them to the corresponding feature extraction branch. Next, the contrast regularization method is utilized to add additional constraints to each branch, thus increasing the separation of target–background information. Furthermore, the self-guided network is proposed to guide the learning of the main branch feature by exploiting texture and topological information in images.

utilizes the texture features and image morphological features of the samples to guide the learning of the main branch to reduce the impact of intraclass differences. Compared with previous methods, MGSNet has the advantage of not blindly processing the entire image uniformly but using different feature extraction branches to simultaneously extract the target, background, and overall features, which is not considered by previous methods. In addition, a guidance mechanism based on density and structural information is designed for the feature extraction of the main branch, so that the network model can better focus on key areas, improving the discriminability and understanding of the model.

In summary, the main contributions are summarized as follows.

1) Aiming at mitigating the impact of background–target imbalance changes and improving the discrimination between similar samples, innovatively separating the target from the background and extracting relevant features in the remote-sensing scene classification task, which is a way that has not been involved in previous work. To this end, a target–background separation strategy is designed to reduce the confusion between similar targets by extracting the key targets and important background information in the scene images separately. Meanwhile, the separation between background and target features is improved by CR, which enables different branches to extract specific information.
2) To control the variability between intraclass samples and capture class-intrinsic features, a self-guided network is developed. Different from ordinary convolutional networks, it guides the collection and learning of features through the inherent texture and morphological features within the samples, which reduces the interference introduced in the process of sample collection and makes the feature representation between intraclass samples more unified.
3) Different from the previous methods, the proposed MGSNet not only solves the well-known problems of interclass target similarity and intraclass image differences in scene classification tasks, but also attempts to introduce background information as a key supplement to the final classification features.

The rest of this article is organized as follows. Section II describes the proposed MGSNet in detail, including how to separate background and target features in the scene and the architecture of the self-guided network. In Section III, extensive experiments and discussions are deployed to verify the effectiveness of the proposed method and the role played by each module. Finally, Section IV draws the conclusion.

II. Proposed MGSNet Framework

The framework of the proposed MGSNet is illustrated in Fig. 2. First, the input image is separated into the background and the target, after which it is sent to the corresponding branches to extract meaningful features. In the gap of feature extraction in multiple stages, the target and background features are separated by performing CR, so that the main branch fully extracts the overall feature of images while converging to the key target features. In addition, the texture and dense maps obtained by processing the input images are used to learn the texture and morphological features contained in the scenes, thus guiding the main branching features to better focus on the unique properties within classes and reduce the impact of sample differences.

A. Target–Background Separation Strategy

Remote-sensing scene labels are usually determined by the internal key targets, so most previous methods try to extract the

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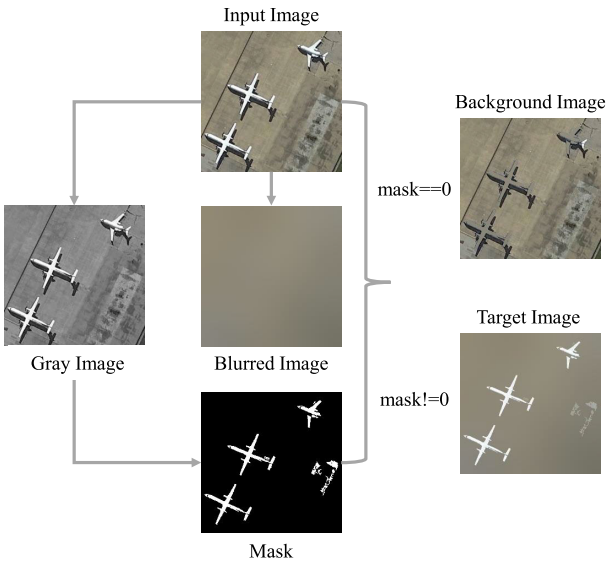


Fig. 3. Illustration of target–background separation strategy.

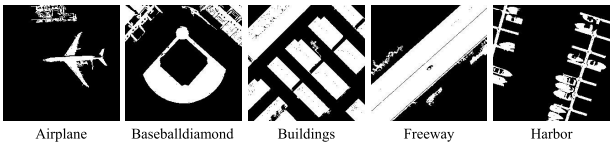


Fig. 4. Segmentation results of images from different classes.

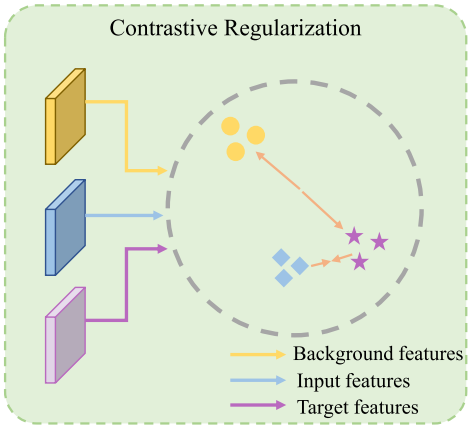


Fig. 5. CR is adopted to pull the main branch features to the target features and push the background features away from the target features.

key target information from the input image. However, actual scenes also contain some regions and targets unrelated to the scene label, which are called the background here. As shown in Fig. 1(a), due to the target similarity between scenes, the key target features between different classes are prone to confusion, while the difference in background information provides decision assistance for sample distinction, thus enhancing the discriminative of interclass samples with similar targets. From the samples, as shown in Fig. 1(b), it can be seen that the background information alone also has interclass similarity problems, and at this time, the difference of key targets provides sufficient information for class judgment. Previous related works try to separate the target from the background and use contextual information; however, their directions are mostly focused on segmentation and detection [34], [35], [36], and there is no complete related work in the field of remote-sensing scene classification before. To this end, a target–background separation strategy is proposed to extract the background and key target information contained in the input image, respectively. The whole process is illustrated in Fig. 3.

To obtain the effective target and background information in the scene, respectively, a corresponding mask map is designed first, which can be calculated as follows:

$$M = \text{Adapthres}(X_{\text{gray}}) \tag{1}$$

where Adapthres is the adaptive threshold segmentation function and X_{gray} is the grayscale map converted from the original image. To better demonstrate the segmentation results, several images from different classes are selected and their segmented results are shown in Fig. 4.

After that, the original image X and the blurred image X_{blurred} obtained by Gaussian blurring are filled in the corresponding area according to the segmentation result, to avoid the influence of other information on the feature extraction of specific branches and make the background and target branches focus on the corresponding features in the image, respectively,

$$X_{\text{target}} = \begin{cases} X, & M \neq 0 \\ X_{\text{blurred}}, & M == 0 \end{cases}$$
$$X_{\text{background}} = \begin{cases} X_{\text{blurred}}, & M \neq 0 \\ X, & M == 0 \end{cases} \tag{2}$$

where X_{target} and $X_{\text{background}}$ are the inputs of the target and background branches.

B. Contrastive Regularization

Inspired by contrastive learning, it aims to learn a feature representation in a certain feature space that draws the "positive" pairs closer and pulls the "negative" pairs apart. To this end, a CR is introduced to normalize and generate better feature representations, which is shown in Fig. 5. There are three feature extraction branches in MGSNet, the background and target branches are used to extract the separated background and target features, and the main branch is used to mine the overall features of the scene image. To this end, the positive and negative pairs in CR are features extracted from the target F_t and main branches F_m and features from the background F_b and target branches, respectively. Therefore, the loss function of the network can be reformulated as follows:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} -y_i^c \log\left(p_i^c\right) + \beta\rho(F_b, F_t, F_m) \tag{3}$$

where the first term is the cross-entropy loss between the true label and the output of MGSNet to regularize the learning of the classifier. N is the number of input samples, C is the number of scene classes, and y is the corresponding label; if the class of input is c, then $y^c = 1$, otherwise 0; p^c is the prediction probability after softmax. The second term

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$\rho(F_b, F_t, F_m)$ represents the CR of F_b, F_t, and F_m under the same latent feature space, which plays an important role in bringing the main branch features closer to the target features and the background features far away the target features. β is a hyperparameter used to balance the cross-entropy loss and CR. To enhance the contrastive power between features, hidden features from the multistage are extracted and subjected to CR. Thus, the overall loss function can be further calculated as follows:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} -y_i^c \log\left(p_i^c\right) + \beta \sum_{j=1}^{h} w_j \frac{D\left(F_m^j, F_t^j\right)}{D\left(F_b^j, F_t^j\right)} \tag{4}$$

where h is the number of hidden layers, and $j = 1, 2, \ldots, h$ is the jth hidden layer from the model. $D(x, y)$ is the L1 loss between x and y, and w_j is the weight coefficient.

C. Self-Guided Network

As mentioned before, due to the varied lighting conditions, view angle, and complexity of ground surfaces, there are still differences in presentation forms between scenes of the same class. To this end, the designed self-guided network reduces the impact of performance differences by exploiting the inherent structural features of samples to guide feature extraction and learning. It consists of two parts, dense map guidance and texture structure guidance, starting from morphological features and texture features, respectively, to improve the intraclass consistency of sample features.

1) Dense Map Guidance: In general, scene images contain complex regions with dense semantic information and smooth regions with low information frequency, and the mixing of these regions poses a challenge for the model to focus on the key semantic regions in the image. Therefore, it is suboptimal to blindly process the entire input sample in the same way. Inspired by the success of noise map in denoising task, a dense map X_D is introduced here to guide the network to perform targeted processing on different regions of the scene. In the dense map, regions with dense semantic information correspond to high-frequency patterns, while regions with less texture correspond to low-frequency patterns. It can be calculated as follows:

$$X_D = g_3(g_2(I_{\text{gray}} - g_1(I_{\text{gray}}; K_1) : K_2)) \tag{5}$$

where both g_1 and g_2 are Gaussian blur operations with the kernel size of K_1 and K_2, g_3 is a normalization function, and I_{gray} is the mean of three channels of the input image. g_3 is defined as follows:

$$g_3(I) = \frac{I - \min(I)}{\max(I) - \min(I) + \varepsilon}. \tag{6}$$

After applying Gaussian blur operation g_1 to the input image, the regions with dense semantic information become blurred, which makes the gap between the output and the original input more obvious. After that, another Gaussian blurring operation g_2 is applied to obtain a smooth density map. Finally, the image is regularized to the interval from 0 to 1. The whole process is shown in Fig. 6.

Once the dense map is computed, similar to [37], it is combined by concatenating it with other channels and then fed into the network as the input to the main branch.

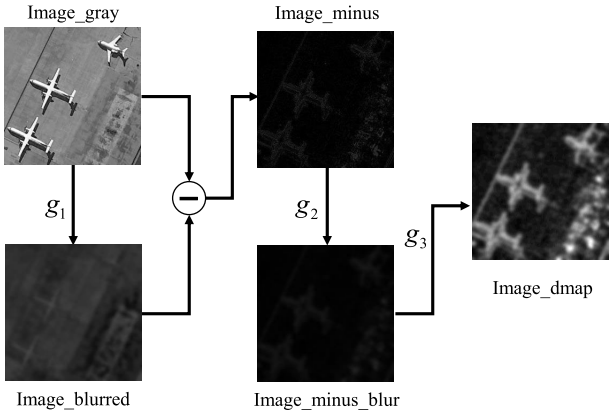


Fig. 6. Illustration of computing the dense map.

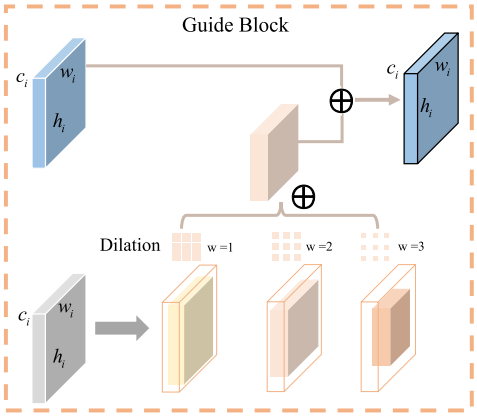


Fig. 7. Guide block with different dilation rates for texture structure guidance.

2) Texture Structure Guidance: Due to the different acquisition times and methods of intraclass samples, their representation forms are diverse, while the inherent texture structure features in the data provide a good guide for feature extraction. For this reason, a texture structure guidance module is proposed, and its structure is shown in Fig. 7.

The gray image obtained from the original input discards the influence caused by color and light intensity and focuses on the texture structure information. The abstract features extracted by different stages are sent to the texture structure guidance module together with the main branch features. Within the module, the features with different receptive fields are obtained through convolution operations with different dilation rates, while effectively modeling the scale variation associated with the object texture. Specifically, let F_m and F_d denote the features extracted by the main branch and the texture features at the same level, respectively. By defining different dilation rates $d = 1, 2, 3$, F_d can obtain features with different receptive fields. Combining these features with main branch features can better model scene features of different scales. This process can be expressed as follows:

$$F_m = F_m \oplus \sum_{d} \text{Conv}(F_d, d). \tag{7}$$

With the guidance of multiscale texture features, the feature extraction and learning of the main branch better deal with the

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interclass samples with different representations and improve the unity of feature representation.

D. Analysis on the Proposed Method

This article proposes a target–background separation strategy and CR for the insufficient feature representation caused by the difference of target–background imbalance changes in scene classification tasks. In addition, in the face of the problem of intraclass sample variability, the self-guided network is used to guide the feature learning of the network branch.

The motivation of this method is to use the background information ignored by existing methods as an aid to scene classification, to improve the discrimination of the model for input samples. To achieve this goal and reduce the interference of irrelevant information, a unique separation strategy is proposed and verified. Only a suboptimal separation can be achieved by processing the input samples alone. Therefore, a CR method is introduced to reduce the interference of irrelevant information under the specific branch, and its enhancement for model understanding is discussed in Section III-D.

Furthermore, to improve the grasp of the overall information of the scene and eliminate the interference generated during the sample collection process, a self-guided network came into being. Its goal is to utilize the uniform texture and topological features of the intraclass samples to compensate for the differences in representation forms and guide the learning of main branch features. Specifically, the combination of the dense map gives extra attention to the key area of the image, and the texture structure information at different scales is captured through the convolution operation with different dilation rates. The combination of them effectively promotes the mining of the essential features of the image.

III. Experimental Results and Discussions

A. Datasets Descriptions and Evaluation Metrics

1) Datasets Descriptions:

1) A series of experiments are performed on the UCM dataset [54], which is one of the most classic datasets in remote-sensing image scene classification tasks. It consists of 21 scene classes, including agricultural, airplane, baseball diamond, beach, and so on. It is derived from the United States Geological Survey National Map, Urban Area Imagery collection, where each class consists of 100 images with a size of 256 × 256 pixels, which together form a scene classification dataset containing 2100 images. Fig. 8 shows some samples of this benchmark dataset. To achieve a fair comparison, 80% of the images are randomly selected as the training set, and the remaining images are used for testing.
2) The AID dataset is used in the experiment [38], which is a large-scale dataset consisting of 10 000 aerial scenes images with a size of 600 × 600 pixels, containing 30 scene categories such as airport, bare land, baseball field, beach, bridge, and so on. All images were collected via Google Earth and annotated by experts, with spatial resolution changes from 0.5 to 8 m. Fig. 9 illustrates some images of each class in this dataset. Similar to UCM, 20% and 50% of the samples are randomly selected as the training set to optimize the model parameters.
3) The NWPU-RESISC45 dataset is a large-scale dataset created by Northwestern Polytechnic University using Google Earth Imagery [12], which contains 45 scene categories including airplane, airport, baseball diamond, basketball court, beach, and so on. Each class consists of 700 images with a size of 256 × 256, and its spatial resolution ranges from 0.2 to 30 m. Fig. 10 exhibits some images of this challenging dataset. 10% and 20% of the images are selected as the training set, and the remaining images are used to test the performance of the model.



Fig. 8. Sample images of the UCM dataset: two images of each class are exhibited (semantic category × number of samples).

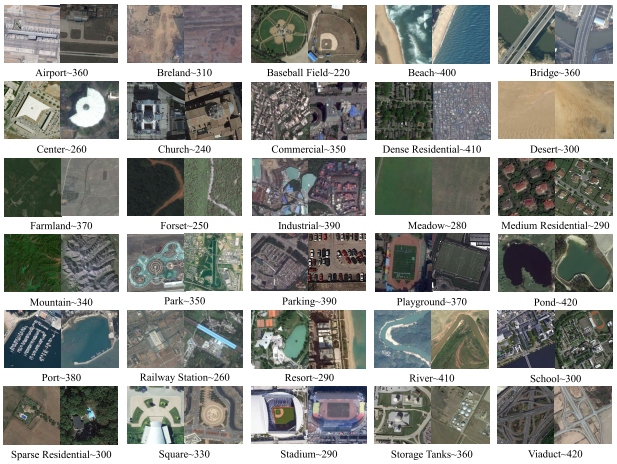


Fig. 9. Sample images of the AID dataset: two images of each class are exhibited (semantic category × number of samples).

2) Evaluation Metrics: For better quantitative analysis of the experimental results, two widely used evaluation metrics were introduced: 1) *Overall Accuracy (OA):* the OA is the ratio between the number of correctly classified samples and the overall samples; 2) *Average Accuracy (AA):* the AA refers to the average of each class accuracy, which reflects the balance of the classification results; and 3) *Kappa Coefficient (KC):* KC an indicator for consistency detection, which is used to measure the effectiveness of the classification. In addition,

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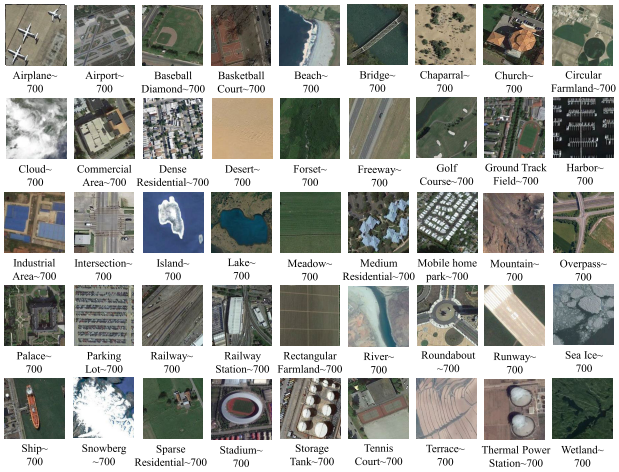


Fig. 10. Sample images of the NWPU-RESISC45 dataset: one image of each class is exhibited (semantic category × number of samples).

the confusion matrix (CM) is introduced to visualize the interclass classification error and the degree of confusion, thus enabling qualitative analysis of the classification results. Specifically, each column represents the prediction result of the model, the sum of each column represents the number of samples predicted to be that class, each row represents the true distribution, and the sum of the data in each row represents the true number of samples in that class. In this way, it shows the correct classification and the misclassification of the model, which can help readers better understand the performance of the model.

B. Comparison With State-of-the-Art Approaches

In this section, the experimental results of the three datasets are described and discussed in detail. To verify the effectiveness of the proposed method, some state-of-the-art methods from recent years are used for comparison. The comparison results on UCM, AID, and NWPU-RESISC45 datasets are reported in Tables I–III, respectively.

Table I lists the classification results of the proposed method and other state-of-the-art methods on the UCM dataset. To prevent the undue influence of the experimental samples on the experimental results and to maintain the fairness of the results of different methods, all methods take 80% of the samples as the training set for model optimization. Judging from the classification accuracy of each method in the table, the understanding ability of handcrafted feature-based methods limits the classification accuracy of the model due to the limitation of feature extraction, thus affecting the final classification results. While the classification accuracy of most deep feature-based methods can reach 90% and then tend to be stable and difficult to improve. MGSNet achieves a higher classification accuracy of 99%, which is a significant improvement compared to other methods. This further demonstrates that the proposed MGSNet not only effectively alleviates the difference caused by the target–background imbalance change, but also improves the feature consistency of the intraclass samples, which is conducive to the improvement of classification accuracy.

TABLE I
COMPARISON OF OA AND STANDARD DEVIATIONS (%) OF STATE-OF-THE-ART METHODS ON THE UCM DATASET WITH THE TRAINING RATIO OF 80%

Type	Method	Publication Year	Training ratio 80% (20% testing)
†	BoVW(LBP) [38]	TGRS2017	77.12±1.93
	BoVW(SIFT) [38]	TGRS2017	74.12±3.30
	salM^{3}LBP-CLM [39]	JSTARS2017	95.75±0.80
	salCLM(eSIFT) [39]	JSTARS2017	94.52±0.79
‡	Two-Fusion [40]	CIN2018	98.02±1.03
	CCPNet [41]	RS2018	97.52±0.97
	GCFs+LOFs [42]	RS2018	99.00±0.35
	CNN-CapsNet [43]	RS2019	99.05±0.24
	SCCov [44]	TNNLS2019	99.05±0.25
	ARCNet-VGG [45]	TGRS2019	99.12±0.40
	GBNet [46]	TGRS2020	98.57±0.48
	MG-CAP [47]	TIP2020	99.00±0.10
	BiMobileNet [48]	Sensors2020	99.03±0.28
	SEMSDNet [49]	JSTARS2021	99.41±0.41
	CSDS [50]	JSTARS2021	99.52±0.13
	T-CNN [51]	TGRS2022	99.33±0.11
	DFAGCN [13]	TNNLS2022	98.48±0.42
Ours	MGSNet		OA: **99.76±0.14** AA: 99.76±0.14 KC: 99.75±0.16

†:Handcrafted Feature-Based Methods ‡:Deep Feature-Based Methods

The classification performance of the comparison methods and MGSNet on the AID dataset are listed in Table II, where two columns of results represent using 20% and 50% samples for training, and the remaining samples as the test set. Compared with handcrafted feature-based methods, deep feature-based methods have an improvement of at least 1.83% and 3.34% when the proportion of training samples is 20% and 50%, with a great improvement. The improvement also fully demonstrates the benefit of deep feature extraction. In addition, with the increase in the number of training samples, the classification accuracy has a greater gap, which indicates the demand for training samples of deep feature-based methods. The proposed MGSNet achieves the best classification performance of 95.46% and 97.18%, respectively, under different training ratios and has improved the accuracy of 8.54%, 7.42%, and 0.63%, 0.31% compared to the two types of methods, which confirms the effectiveness of the proposed method on the AID dataset.

Compared with the first two datasets, the NWPU-RESISC45 dataset has the largest number of scene classes as well as the amount of data, so its rich image variations, differences of intraclass samples, and similarities between interclass samples make the dataset more challenging. The comparison results between the proposed method and existing methods using the NWPU-RESISC45 dataset are listed in Table III. With 10% and 20% of the samples are, respectively, chosen as the training set, and the remaining samples are used for testing, the proposed method outperforms other comparative methods. When the training rate is 10%, the OA of MGSNet is 2.15% higher than the latest research T-CNN based on depth feature, and 50.68% higher than the methods based on handcrafted

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TABLE II
COMPARISON OF OA AND STANDARD DEVIATIONS (%) OF STATE-OF-THE-ART METHODS ON THE AID DATASET WITH THE TRAINING RATIO OF 20% AND 50%

Type	Method	Publication Year	Training ratios	
			20%(80% testing)	50%(50% testing)
†	BoVW(LBP)) [38]	TGRS2017	56.73±0.54	64.25±0.55
	BoVW(SIFT) [38]	TGRS2017	61.40±0.41	67.65±0.49
	salM^{3}LBP-CLM [39]	JSTARS2017	86.92±0.35	89.76±0.45
	salCLM(eSIFT) [39]	JSTARS2017	85.58±0.83	88.41±0.63
‡	Two-Fusion [40]	CIN2018	92.32±0.41	94.58±0.25
	GCFs+LOFs [42]	RS2018	92.48±0.38	96.85±0.23
	CNN-CapsNet [43]	RS2019	93.79±0.13	96.32±0.12
	SCCov [44]	TNNLS2019	93.12±0.25	96.10±0.16
	ARCNet-VGG [45]	TGRS2019	88.75±0.40	93.10±0.55
	GBNet [46]	TGRS2020	92.20±0.23	95.48±0.12
	MG-CAP [47]	TIP2020	93.34±0.18	96.12±0.12
	BiMobileNet [48]	Sensors2020	94.83±0.24	96.87±0.23
	CSDS [50]	JSTARS2021	94.29±0.35	96.70±0.14
	PSGAN [52]	TGRS2022	89.47±0.34	92.67±0.55
	T-CNN [51]	TGRS2022	94.55±0.27	96.27±0.23
Ours	MGSNet		OA: **95.46±0.21** AA: 95.25±0.27 KC: 95.60±0.31	OA: **97.18±0.16** AA: 97.12±0.19 KC: 97.08±0.22

†:Handcrafted Feature-Based Methods ‡:Deep Feature-Based Methods

TABLE III
COMPARISON OF OA AND STANDARD DEVIATIONS (%) OF STATE-OF-THE-ART METHODS ON THE NWPU-RESISC45 DATASET WITH THE TRAINING RATIO OF 10% AND 20%

Type	Method	Publication Year	Training ratios	
			10%(90% testing)	20%(80% testing)
†	BoVW [12]	RPOC2017	41.72±0.21	44.79±0.28
	BoVW+SPM [12]	RPOC2017	27.83±0.61	32.96±0.47
	LLC [12]	RPOC2017	38.81±0.23	40.03±0.34
‡	Fine-tuned VGG-16 [12]	RPOC2017	87.15±0.45	90.36±0.18
	Two-Fusion [40]	CIN2018	80.22±0.22	83.16±0.18
	CNN-CapsNet [43]	RS2019	89.03±0.21	92.60±0.11
	SCCov [44]	TNNLS2019	89.30±0.35	92.10±0.25
	MF^{2}Net [53]	GRSL2020	90.17±0.25	92.73±0.21
	MG-CAP [47]	TIP2020	90.83±0.12	92.95±0.11
	BiMobileNet [48]	Sensors2020	92.06±0.14	94.08±0.11
	SEMSDNet [49]	JSTARS2021	91.68±0.39	93.89±0.63
	CSDS [50]	JSTARS2021	91.64±0.16	93.59±0.21
	PSGAN [52]	TGRS2022	84.72±0.72	88.47±0.56
	T-CNN [51]	TGRS2022	90.25±0.14	93.05±0.12
Ours	MGSNet		OA: **92.40±0.16** AA: 92.40±0.16 KC: 92.23±0.18	OA: **94.57±0.12** AA: 94.57±0.12 KC: 94.44±0.15

†:Handcrafted Feature-Based Methods ‡:Deep Feature-Based Methods

feature. With a 20% training share, there is an improvement of 1.52% and 49.78%. It can be seen that the learning ability and discrimination of the model are effectively improved by introducing background information and improving the consistency of intraclass samples.

C. Reports of CM

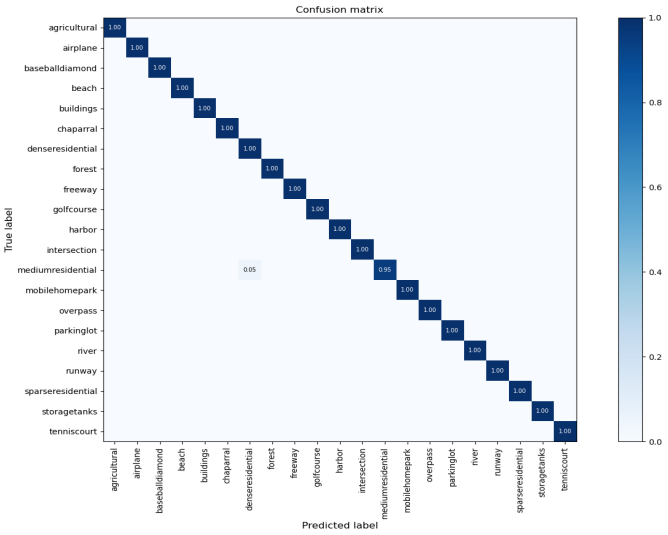
To better demonstrate the specific classification performance of the proposed method, confusion matrices are drawn to illustrate the classification results. A total of five confusion matrices can be obtained through experimental results with different training ratios on the UCM, AID, and NWPU-RESISC45 datasets, which are presented in Figs. 11–15. Each column of the CM represents the predicted label, and the sum of each column represents the number of samples predicted to be the corresponding scene. Each row represents the actual label of the sample, and the sum of each row represents the number of samples of the corresponding scene in the test set. Thus, the cells on the diagonal represent the proportion of samples that are correctly classified, while the other cells are misclassified cases.

Fig. 11 shows the CM when the training set accounted for 80% on the UCM dataset. Among the 21 scene classes, only one scene has an accuracy below 100%, reaching 95%,

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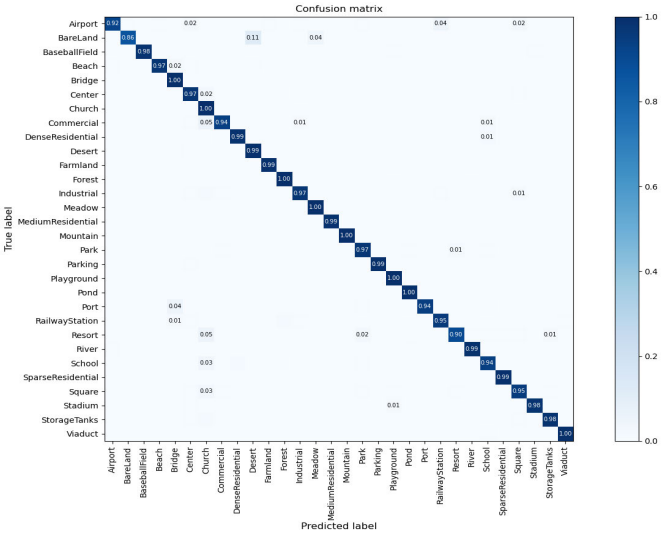
TABLE IV
ABLATION STUDIES FOR THE PROPOSED MGSNET ON THREE DATASETS

Variant	Target-background separation strategy		CR	Self-guided network		OA (%) on different dataset				
	Target branch	Background branch		Dense map guidance	Texture structure guidance	UCM(80%)	AID(20%)	AID(50%)	NWPU(10%)	NWPU(20%)
1	✓	✓	✓	✓	✓	99.76	95.46	97.18	92.40	94.57
2	✓	✓		✓	✓	99.52	95.20	96.92	92.11	94.28
3		✓		✓	✓	99.36	94.98	96.60	91.87	93.99
4	✓			✓	✓	99.28	94.73	96.41	91.66	93.72
5				✓	✓	99.04	94.31	94.18	91.32	93.40
6	✓	✓	✓	✓		99.33	94.72	96.45	91.82	93.91
7	✓	✓	✓		✓	99.56	95.02	96.84	92.04	94.15
8	✓	✓	✓			99.20	94.47	96.21	91.66	93.64
9	✓	✓				99.01	94.23	95.98	91.41	93.39



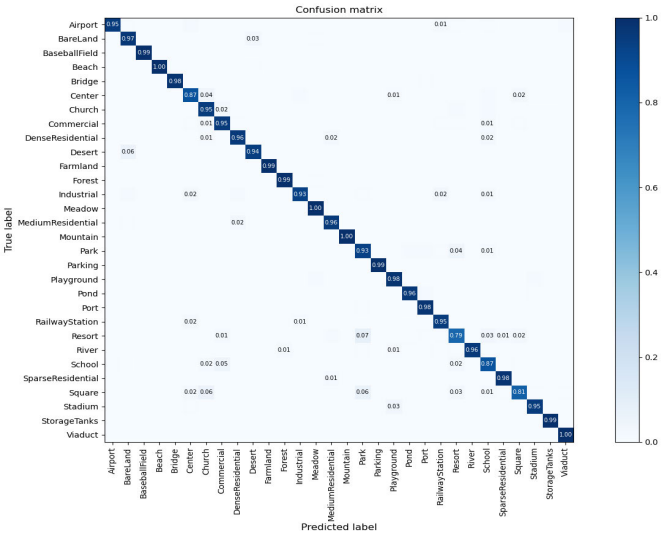
CM of UCM dataset (Training ratio = 80%)

Fig. 11. CM on the UCM dataset with 80% of the dataset for training and the rest for testing.



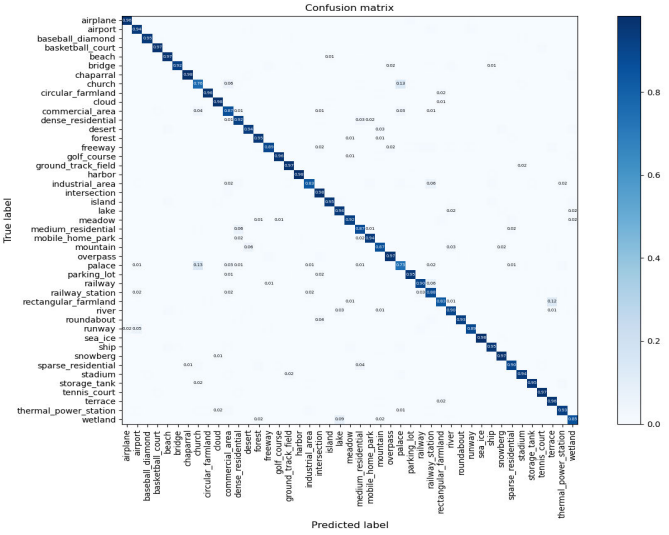
CM of AID dataset (Training ratio = 50%)

Fig. 13. CM on the AID dataset with 50% of the dataset for training and the rest for testing.



CM of AID dataset (Training ratio = 20%)

Fig. 12. CM on the AID dataset with 20% of the dataset for training and the rest for testing.

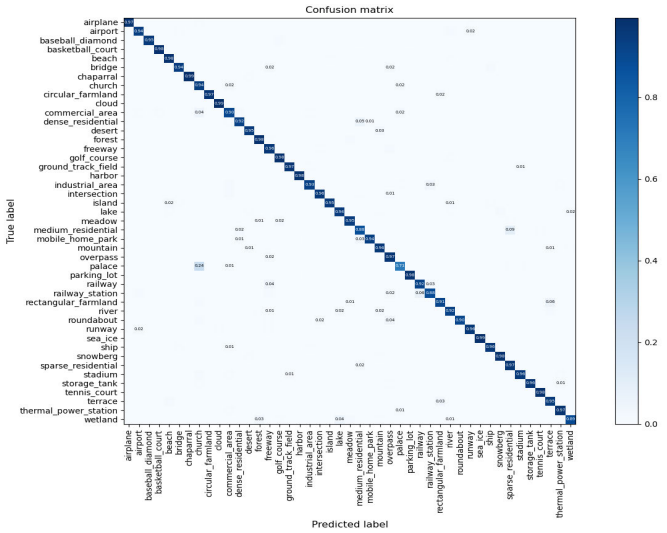


CM of NWPU-RESISC45 dataset (Training ratio = 10%)

Fig. 14. CM on the NWPU-RESISC45 dataset with 10% of the dataset for training and the rest for testing.

which is the misclassification of *medium residential* into *dense residential*. This is because the target and background information in these two types of scenes are too similar, so it is easy to cause model misclassification when processing relevant samples. However, MGSNet still has unique advantages, for example, when dealing with *freeway*, *runway*, and *overpass*,

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CM of NWPU-RESISC45 dataset (Training ratio = 20%)

Fig. 15. CM on the NWPU-RESISC45 dataset with 20% of the dataset for training and the rest for testing.

although the valid targets are all roads, the differences in their background information can be effectively captured by the network to correctly distinguish different classes.

Figs. 12 and 13 show the confusion matrices obtained on the AID dataset when the training set accounts 20% and 50%. Only four and one scene classes' accuracy is less than 90%, and the rest classes have achieved excellent performance. For example, the *center* has more misclassifications because its object shape and context information are similar to the *church*. However, the problem of intraclass variation can be better resolved in the division of classes such as *forest* due to the introduction of texture and morphological information for the guidance of network learning.

The confusion matrices of the NWPU-RESISC45 dataset are shown in Figs. 14 and 15. When the proportion of training samples is 10% and 20%, there are 34 and 41 scenes with classification accuracy exceeding 90%. Among the two groups of experiments, the worst classification performance is *palace*, which is wrongly classified as *church* in most cases. This is because there are certain similarities in valid targets, background, and context information, which makes the model prone to misclassification. On the contrary, although *desert* has a certain similarity with *lake* in the target performance, MGSNet can recognize it well with the assistance of background information. Similarly, the similarity between *chaparral* and *desert* backgrounds can be effectively distinguished by supplementing the information of valid targets, thereby improving the robustness of the model.

D. Ablation Studies

To verify the role of separation strategy, CR, and self-guided network in this task, this section conducts a series of ablation experiments to explore the importance of target–background information for network performance improvement and the effectiveness of various parts of the self-guided network. The specific experimental results are listed in Table IV. Specifically, the whole ablation experiment can be divided into two parts.

1) Verify the effect of target–background information and CR on improving feature discrimination. Through the comparison of 1 and 2, the introduction of CR increases the separation between different features and improves the extraction ability of specific features by each network branch. And the comparisons between 2, 3, 4, and 5 fully demonstrate that the additional introduction of target–background information helps improve the discrimination between similar scenes for better interclass sample differentiation. While 9 fully demonstrates the experimental performance only using the variant with the target branch and background branch. It can be seen that even without adding other modules, good results can still be achieved through the separation and support of the target and background information.
2) Verify the assistance of a self-guided network for main branch feature extraction. Through the comparison of the results of 1, 6, 7, and 8, it can clearly see the benefits of dense graph guidance and texture structure guidance for improving the consistency of the main branch feature representation. They utilize the morphology and texture features of images to guide the feature attention, reducing the impact of differences in intraclass sample representations (e.g., seasonal turnover and regional variation), and allowing the model to focus more on common features within the samples. By combining the target–background separation strategy, CR, and self-guided network, the proposed MGSNet effectively alleviates the misclassification phenomenon caused by the interclass similarity and the intraclass differences, making it achieve state-of-the-art performance in multiple datasets.

E. Running Time and Memory Requirement

To demonstrate the running efficiency and memory requirement of the model, relevant statistics are performed on the UCM dataset. The training time of the model is calculated with 32 images within a round of training, while the test time is obtained using the entire test set (420 images). The training time is 0.307 s and the testing time is 0.614 s, which is the average of the five runs of the model. In addition, the memory requirement of the model is 7.58 MB. From the obtained results, the model runs efficiently and requires a moderate amount of memory, which is sufficient for subsequent applications.

IV. Conclusion

In this article, a novel MGSNet has been proposed for remote-sensing scene classification. First of all, to solve the problem of interclass sample similarity caused by the unbalanced change of target–background, a target–background separation strategy and a CR method are established in MGSNet. Specifically, the input samples are divided into target and background regions by separating them. Afterward, they are fed into the corresponding feature extraction branches to

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achieve separate extraction of target and background information, thus providing more and more reliable information assistance for sample differentiation. And CR serves this strategy, improving the feature separation by expanding the difference between them so that each branch pays more attention to the extraction and utilization of its corresponding information. In addition, a self-guided network based on image texture and morphological features is proposed to cope with differentiated intraclass samples. With the guidance of dense map and texture structure information, the main branch pays more attention to the common characteristics of samples in the class. Extensive experimental results on three datasets demonstrate the superiority of MGSNet from the perspective of quantitative indicators and CM, proving its effectiveness on remote-sensing scene classification tasks.

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